
UNIT 5 PHYSICAL AND SOCIAL INFRASTRUCTURE

Structure

- 5.0 Objectives
- 5.1 Infrastructure in India
 - 5.1.1 Meaning and Significance
 - 5.1.2 Nature of Infrastructure
 - 5.1.3 Types of Infrastructure
 - 5.1.4 Infrastructure and Economic Growth
 - 5.1.5 Role of Public Private Partnership in Infrastructure
 - 5.1.6 Budget Allocations for Infrastructure Sector
 - 5.1.7 Social and Physical Infrastructure
- 5.2 Privatisation and Commercialisation of Infrastructure
 - 5.2.1 Need for Privatisation and Commercialisation
 - 5.2.2 Prerequisites for Private Investment
- 5.3 Physical Infrastructure: Growth and Policy Issues
 - 5.3.1 Transport
 - 5.3.1.1 Road
 - 5.3.1.2 Railways
 - 5.3.1.3 Shipping and Ports
 - 5.3.1.4 Aviation
 - 5.3.2 Telecommunications
 - 5.3.3 Power
 - 5.3.4 Banking
 - 5.3.5 Housing and Urban Infrastructure
- 5.4 Social Infrastructure: Growth and Policy Issues
 - 5.4.1 Education
 - 5.4.2 Health
- 5.5 Infrastructure: Challenges and Way Ahead
- 5.6 Let Us Sum Up
- 5.7 Key Words
- 5.8 References
- 5.9 Answers or Hints to Check Your Progress Exercises

5.0 OBJECTIVES

After reading this unit, you will be able to:

- state the need for infrastructural development in India;
- distinguish between physical and social infrastructure sector;
- explain why both physical and social infrastructure are crucial for India's growth;
- identify the issues requiring policy attention arising out of India's economic growth; and

- pinpoint the issues crucial in infrastructure policy in the Indian economy.

5.1 INFRASTRUCTURE IN INDIA

In Economic literature, infrastructure is popular by the name “Overhead Capital” or “Social Overhead Capital”. The famous economist A.O Hirschman stated that Social Overhead capital is the “basic services without which primary, secondary and tertiary productive activities cannot function”. Importance of infrastructure in facilitating economic and social activity cannot be under-emphasised. Sustainable Development Goal (SDG) 9 aims to “Develop quality, reliable, sustainable and resilient infrastructure, including regional and trans-border infrastructure, to support economic development and human well-being, with a focus on affordable and equitable access for all”.

This unit focuses on both the physical and social infrastructure sector; its importance in the economic growth, recent trends and government policies towards infrastructural development and challenges faced by the sector.

5.1.1 Meaning and Significance

Infrastructure means those basic services and support systems that facilitate different economic activities, thereby helping in the development of a country. Infrastructure comprises of activities such as (i) transport, (ii) communications, (iii) energy, (iv) aspects related to economic activity that contribute to increase in productivity of natural resources, such as irrigation, drainage, afforestation, etc., (vi) science and technology, (vii) information system, (viii) finance and banking, (ix) civic amenities like piped water supply, sanitation and sewerage, solid waste collection and disposal, and piped gas, and (x) human resource services like education and health. Core Infrastructure incorporates all the main types of infrastructure, for instance, roads, highways, railways, public transportation, water, and gas supply, etc. Core assets provide essential services and have monopolistic characteristics. It is useful to view infrastructure as physical infrastructure like roads or bridges and the infrastructure services like transportation, in this case. Most infrastructure services and part of physical infrastructure are non-tradable, and therefore their availability cannot be readily augmented through imports. They need to be domestically produced. Investors seeking to invest in core infrastructure look for five different characteristics: income, low volatility of returns, diversification, inflation protection, and long-term liability matching.

5.1.2 Nature of Infrastructure

The infrastructure sector has certain peculiarities that help us to distinguish this sector from other sectors of the economy. Among these, the more important distinguishing features can be identified as follows:

- Public Goods:** Most of the physical infrastructure services have some elements of public good in them. These services are available to the public; the consumers may be charged for these services or the same may be supplied free. But even when they are supplied against a price, it is not always possible to exclude those consumers who chose not to pay for them.
- Externalities:** The social benefit of the infrastructure services far exceeds the cost involved in their generation. This, in turn, creates problems in pricing

of these services. It makes it difficult to price them in order to recover the cost fully.

- iii) **Monopolies:** Due to the inherent nature of infrastructure services, it is difficult for more than one supplier to exist in one location. It thus creates the possibility for monopolies and their regulation.
- iv) **Public Sector Domination:** The existence of externalities particularly in the field of social welfare has resulted in a dominant position by public sector in production and supply of infrastructure services.
- v) **Lumpy Investment:** Infrastructure prospects, more generally, require lump sum investment, i.e., any expenditure on a part of the project is not useful until the whole project is ready for operations. A half-way road or a rail line are of no use. Similarly, investment in energy is useful only when the whole system of generation, distribution and its reach to the ultimate consumer is ready.
- vi) **Indivisibilities:** Lump-sum investment, in turn, is the result of indivisibilities, a characteristic feature of most infrastructure projects. One cannot divide and sub-divide such projects in small parts and activate them. These are indivisible.

Infrastructural facilities are very necessary and vital for the smooth functioning of the economy. According to Dr V. K. R. V. Rao, “The link between infrastructure and development is not a once for all affair. It is a continuous process and progress in development has to be preceded, accompanied and followed by progress in infrastructure; if we are to fulfil our declared objectives of a self-accelerating process of economic development”.

5.1.3 Types of Infrastructure

Infrastructure can be put into several different types including:

- **Soft infrastructure:** These types of infrastructure make up institutions that help maintain the economy. These usually require human capital and help deliver certain services to the population. Examples include the healthcare system, financial institutions, governmental systems, law enforcement, and education systems. Soft infrastructure refers to all the institutions that maintain the economic, health, social, and cultural standards of a country. This includes educational programmes, official statistics, parks and recreational facilities, law enforcement agencies, and emergency services.
- **Hard Infrastructure:** These make up the physical systems necessary to run a modern, industrialised nation. Examples include roads, highways, bridges, as well as the capital/assets needed to make them operational (transit buses, vehicles, oil rigs/refineries). The assets constituting the hard infrastructure support multiple services that are important intermediaries in the production process. Thus, roads support transportation services that are important to move inputs and outputs of a production process.
- **Critical Infrastructure:** These are assets defined by a government as being essential to the functioning of a society and economy, such as facilities for shelter and heating, telecommunication, public health, agriculture, or civic amenities, etc.